

JavaScript Data Types

[Back to JS page](#)

Table of Contents

- [JavaScript Data Types](#)
 - [JavaScript has 8 Datatypes](#)
 - [Example 1](#)
 - [The typeof Operator](#)
 - [Example 2](#)
 - [JavaScript Strings](#)
 - [Example 3](#)
 - [JavaScript Numbers](#)
 - [Example 4](#)
 - [Exponential Notation](#)
 - [Example 5](#)
 - [JavaScript Booleans](#)
 - [Example 6](#)
 - [Datatype undefined](#)
 - [Example 7](#)
 - [Empty Values](#)
 - [Example 8](#)
 - [Document](#)
 - [Reference](#)

JavaScript has 8 Datatypes

JavaScript has 8 datatypes:

1. **String**
2. **Number**
3. **Bigint**
4. **Boolean**
5. **Undefined**
6. **Null**
7. **Symbol**
8. **Object**

Objects can be: An object, an array, or a date.

```
<!DOCTYPE html>
<html>
<body>

<h2>JavaScript Data Types</h2>
<p>From w3schools.com, Experiment by Teeratus_R</p>

<h4>JavaScript has 8 Datatypes</h4>
<p id="demo1"></p>
<p id="demo2"></p>
<p id="demo3"></p>
<p id="demo4"></p>
<p id="demo5"></p>

<script>
// Numbers:
let length = 16;
let weight = 7.5;

// Strings:
let color = "Yellow";
let lastName = "Johnson";

// Booleans:
let x = true;
let y = false;

// Object:
const person = {firstName:"John", lastName:"Doe"};

// Array object:
const cars = ["Saab", "Volvo", "BMW"];

// Date object:
const date = new Date("2022-03-25");

document.getElementById("demo1").innerHTML = "Number: " + length + ", " + weight;
document.getElementById("demo2").innerHTML = "String: " + color + ", " + lastName;
document.getElementById("demo3").innerHTML = "Boolean: " + x + ", " + y;
document.getElementById("demo4").innerHTML = "Object: " + person.firstName + " " + person.lastName;
document.getElementById("demo5").innerHTML = "Array: " + cars;
</script>

</body>
</html>
```

Example 1

Result [View Example](#)

The typeof Operator

You can use the `typeof` operator to find the data type of a JavaScript variable:

```
<!DOCTYPE html>
<html>
<body>

<h2>JavaScript Data Types</h2>
<p>From w3schools.com, Experiment by Teeratus_R</p>

<h4>The typeof Operator</h4>
<p id="demo"></p>

<script>
document.getElementById("demo").innerHTML =
  "typeof 'John' is: " + typeof "John" + "<br>" +
  "typeof 3.14 is: " + typeof 3.14 + "<br>" +
  "typeof true is: " + typeof true + "<br>" +
  "typeof x is: " + typeof x + "<br>" +
  "typeof [1,2,3] is: " + typeof [1,2,3] + "<br>" +
  "typeof {name:'John'} is: " + typeof {name:"John"};
</script>

</body>
</html>
```

Example 2

Result [View Example](#)

JavaScript Strings

A string (or a text string) is a series of characters like "John Doe".

Strings are written with quotes. You can use single or double quotes:

```
<!DOCTYPE html>
<html>
<body>

<h2>JavaScript Data Types</h2>
<p>From w3schools.com, Experiment by Teeratus_R</p>

<h4>JavaScript Strings</h4>
<p id="demo"></p>

<script>
let carName1 = "Volvo XC60";
let carName2 = 'Volvo XC60';
document.getElementById("demo").innerHTML =
  carName1 + "<br>" +
  carName2;
</script>

</body>
</html>
```

Example 3

Result [View Example](#)

JavaScript Numbers

All JavaScript numbers are stored as decimal numbers. Floating point arithmetic is not always 100% accurate:

```

<!DOCTYPE html>
<html>
<body>

<h2>JavaScript Data Types</h2>
<p>From w3schools.com, Experiment by Teeratus_R</p>

<h4>JavaScript Numbers</h4>
<p id="demo1"></p>
<p id="demo2"></p>

<script>
let x1 = 34.00;
let x2 = 34;
let x3 = 3.14;

document.getElementById("demo1").innerHTML =
  "x1 = " + x1 + "<br>" +
  "x2 = " + x2 + "<br>" +
  "x3 = " + x3;

let y1 = 123e5;
let y2 = 123e-5;
document.getElementById("demo2").innerHTML =
  "123e5 = " + y1 + "<br>" +
  "123e-5 = " + y2;
</script>

</body>
</html>

```

Example 4

Result [View Example](#)

Exponential Notation

Extra large or extra small numbers can be written with scientific (exponential) notation:

```
<!DOCTYPE html>
<html>
<body>

<h2>JavaScript Data Types</h2>
<p>From w3schools.com, Experiment by Teeratus_R</p>

<h4>Exponential Notation</h4>
<p id="demo"></p>

<script>
let y1 = 123e5;
let y2 = 123e-5;
document.getElementById("demo").innerHTML =
  "123e5 = " + y1 + " (large number)" + "<br>" +
  "123e-5 = " + y2 + " (small number)";
</script>

</body>
</html>
```

Example 5

Result [View Example](#)

JavaScript Booleans

Booleans can only have two values: `true` or `false` :

```
<!DOCTYPE html>
<html>
<body>

<h2>JavaScript Data Types</h2>
<p>From w3schools.com, Experiment by Teeratus_R</p>

<h4>JavaScript Booleans</h4>
<p id="demo"></p>

<script>
let x = 5;
let y = 5;
let z = 6;
document.getElementById("demo").innerHTML =
  "(x == y) = " + (x == y) + "<br>" +
  "(x == z) = " + (x == z);
</script>

</body>
</html>
```

Example 6

Result [View Example](#)

Datatype undefined

A variable without a value has the value `undefined` . The type is also `undefined` :

```
<!DOCTYPE html>
<html>
<body>

<h2>JavaScript Data Types</h2>
<p>From w3schools.com, Experiment by Teeratus_R</p>

<h4>Datatype undefined</h4>
<p id="demo"></p>

<script>
let car;
document.getElementById("demo").innerHTML =
  "Value: " + car + "<br>" +
  "Type: " + typeof car;
</script>

</body>
</html>
```

Example 7

Result [View Example](#)

Empty Values

An empty value has nothing to do with `undefined` . An empty string has both a legal value and a type:


```
<!DOCTYPE html>
<html>
<body>

<h2>JavaScript Data Types</h2>
<p>From w3schools.com, Experiment by Teeratus_R</p>

<h4>Empty Values</h4>
<p id="demo"></p>

<script>
let car = "";
document.getElementById("demo").innerHTML =
  "Value: '" + car + "'<br>" +
  "Type: " + typeof car;
</script>

</body>
</html>
```

Example 8

Result [View Example](#)

Document

Document in project

You can [Download PDF](#) file.

Reference

- [W3Schools JavaScript Data Types](#)